

Questions

MathonGo

Q1 - 24 June - Shift 1

If $x = x(y)$ is the solution of the differential

equation $y \frac{dx}{dy} = 2x + y^3(y+1)e^y$, $x(1) = 0$; then $x(e)$

is equal to :

(A) $e^3(e^e - 1)$ (B) $e^e(e^3 - 1)$

(C) $e^2(e^e + 1)$ (D) $e^e(e^2 - 1)$

Space for your notes:

Q2 - 24 June - Shift 2

The slope of normal at any point (x, y) , $x > 0$, $y > 0$

on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$.

If the curve passes through the point $(1, 1)$, then

$e.y(e)$ is equal to

(A) $\frac{1 - \tan(1)}{1 + \tan(1)}$ (B) $\tan(1)$

(C) 1 (D) $\frac{1 + \tan(1)}{1 - \tan(1)}$

Space for your notes:

Q3 - 25 June - Shift 1

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Let $y = y(x)$ be the solution of the differential equation $(x + 1)y' - y = e^{3x}(x + 1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve $y = y(x)$ is:

- (A) not a critical point
- (B) a point of local minima
- (C) a point of local maxima
- (D) a point of inflection

*Space for your notes:***Q4 - 25 June - Shift 1**

If the solution curve $y = y(x)$ of the differential equation $y^2 dx + (x^2 - xy + y^2) dy = 0$, which passes through the point $(1, 1)$ and intersects the line $y = \sqrt{3} x$ at the point $(\alpha, \sqrt{3} \alpha)$, then value of $\log_e(\sqrt{3} \alpha)$ is equal to

- (A) $\frac{\pi}{3}$
- (B) $\frac{\pi}{2}$
- (C) $\frac{\pi}{12}$
- (D) $\frac{\pi}{6}$

*Space for your notes:***Q5 - 25 June - Shift 2**

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If $y = y(x)$ is the solution of the differential equation $2x^2 \frac{dy}{dx} - 2xy + 3y^2 = 0$ such that

$y(e) = \frac{e}{3}$, then $y(1)$ is equal to

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) 3

*Space for your notes:***Q6 - 26 June - Shift 1**

Let the solution curve $y = y(x)$ of the differential equation $(4 + x^2)dy - 2x(x^2 + 3y + 4)dx = 0$ pass through the origin. Then $y(2)$ is equal to _____.

*Space for your notes:***Q7 - 26 June - Shift 1**

Let $S = (0, 2\pi) - \left\{ \frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4} \right\}$. Let $y =$

$y(x)$, $x \in S$, be the solution curve of the differential

equation $\frac{dy}{dx} = \frac{1}{1 + \sin 2x}$, $y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum

of abscissas of all the points of intersection of the

curve $y = y(x)$ with the curve $y = \sqrt{2} \sin x$ is $\frac{k\pi}{12}$,

then k is equal to _____.

*Space for your notes:***Q8 - 26 June - Shift 2**

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If the solution of the differential equation

$$\frac{dy}{dx} + e^x(x^2 - 2)y = (x^2 - 2x)(x^2 - 2)e^{2x} \text{ satisfies}$$

$y(0) = 0$, then the value of $y(2)$ is _____.

- (A) -1 (B) 1 (C) 0 (D) e

Space for your notes:

Q9 - 27 June - Shift 1

Let $\frac{dy}{dx} = \frac{ax - by + a}{bx + cy + a}$, where a, b, c are constants,

represent a circle passing through the point (2, 5).

Then the shortest distance of the point (11, 6) from

this circle is :

- (A) 10 (B) 8
(C) 7 (D) 5

Space for your notes:

Q10 - 27 June - Shift 1

If $\frac{dy}{dx} + \frac{2^{x-y}(2^y - 1)}{2^x - 1} = 0$, $x, y > 0$, $y(1) = 1$, then

$y(2)$ is equal to :

- (A) $2 + \log_2 3$ (B) $2 + \log_2 2$
(C) $2 - \log_2 3$ (D) $2 - \log_2 3$

Space for your notes:

Q11 - 27 June - Shift 2

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If the solution curve of the differential equation $((\tan^{-1} y) - x) dy = (1 + y^2) dx$ passes through the point $(1, 0)$ then the abscissa of the point on the curve whose ordinate is $\tan(1)$ is :

- (A) $2e$ (B) $\frac{2}{e}$
(C) 2 (D) $\frac{1}{e}$

Space for your notes:

Q12 - 27 June - Shift 2

Let v be the solution of the differential equation $(1 - x^2) dy = (xy + (x^3 + 2)\sqrt{1 - x^2}) dx$, $-1 < x < 1$

and $y(0) = 0$ if $\int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{1 - x^2} y(x) dx = k$ then k^{-1} is

equal to :

Space for your notes:

Q13 - 28 June - Shift 1

Let the solution curve $y = y(x)$ of the differential equation,

$$\left[\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}} \right] x \frac{dy}{dx} = x + \left[\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}} \right] y$$

pass through the points $(1, 0)$ and $(2\alpha, \alpha)$, $\alpha > 0$.

Then α is equal to

- (A) $\frac{1}{2} \exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$ (B) $\frac{1}{2} \exp\left(\frac{\pi}{3} + \sqrt{e} - 1\right)$
(C) $\exp\left(\frac{\pi}{6} + \sqrt{e} + 1\right)$ (D) $2 \exp\left(\frac{\pi}{3} + \sqrt{e} - 1\right)$

Space for your notes:

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Q14 - 28 June - Shift 1

Let $y = y(x)$ be the solution of the differential equation $x(1-x^2)\frac{dy}{dx} + (3x^2y - y - 4x^3) = 0, x > 1$,

with $y(2) = -2$. Then $y(3)$ is equal to

- (A) -18 (B) -12
(C) -6 (D) -3

Space for your notes:

Q15 - 28 June - Shift 2

Let $x = x(y)$ be the solution of the differential equation $2ye^{x/y^2}dx + (y^2 - 4xe^{x/y^2})dy = 0$ such that $x(1) = 0$. Then, $x(e)$ is equal to

- (A) $e \log_e(2)$ (B) $-e \log_e(2)$
(C) $e^2 \log_e(2)$ (D) $-e^2 \log_e(2)$

Space for your notes:

Q16 - 29 June - Shift 1

Let the solution curve of the differential equation

$$x \frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}, y(1) = 3 \text{ be } y = y(x).$$

Then $y(2)$ is equal to :

- (A) 15 (B) 11
(C) 13 (D) 17

Space for your notes:

Q17 - 29 June - Shift 1

Questions

MathonGo

Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = x e^{\tan^{-1}(\sqrt{2}\cot 2x)}, 0 < x < \frac{\pi}{2} \text{ with } y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}.$$

If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18} e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to _____.

Space for your notes:

Q18 - 29 June - Shift 2

If $y = y(x)$ is the solution of the differential equation $(1 + e^{2x}) \frac{dy}{dx} + 2(1 + y^2)e^x = 0$ and $y(0) = 0$,

then $6\left(y'(0) + \left(y(\log_e \sqrt{3})\right)^2\right)$ is equal to:

(A) 2 (B) -2

(C) -4 (D) -1

Space for your notes:

Q19 - 29 June - Shift 2

Let $y = y(x)$, $x > 1$, be the solution of the differential equation $(x-1) \frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with

$y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha + 1}{\beta e^\alpha}$, then the value of $\alpha + \beta$ is equal to _____.

Space for your notes:

Answer Key

Q1 (A)

Q2 (D)

Q3 (B)

Q4 (C)

Q5 (B)

Q6 (12)

Q7 (42)

Q8 (C)

Q9 (B)

Q10 (D)

Q11 (B)

Q12 (320)

Q13 (A)

Q14 (A)

Q15 (D)

Q16 (A)

Q17 (2)

Q18 (C)

Q19 (14)

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Q1 (A)

$$y \frac{dx}{dy} = 2x + y^3(y+1)e^y, x(1) = 0$$

$$\frac{dx}{dy} - \frac{2}{y}x = y^2(y+1)e^y$$

$$\text{I.f} = e^{\int \frac{-2}{y} dy} = \frac{1}{y^2}$$

$$x \cdot \frac{1}{y^2} = \int (y+1)e^y dy$$

$$\frac{x}{y^2} = (y+1)e^y - e^y + c = y \cdot e^y + c$$

$$x = y^3 e^y + cy^2$$

$$\text{For } x = 0, y = 1 \Rightarrow c = -e$$

$$x = y^3 e^y - e \cdot y^2$$

$$x(e) = e^3(e^e - 1)$$

Q2 (D)

$$\text{Slope of normal} = \frac{-dx}{dy} = \frac{x^2}{xy - x^2y^2 - 1}$$

$$x^2y^2dx + dx - xydx = x^2dy$$

$$x^2y^2dx + dx = x^2dy + xydx$$

$$x^2y^2dx + dx = x(xdy + ydx)$$

$$x^2y^2dx + dx = xd(xy)$$

$$\frac{dx}{x} = \frac{d(xy)}{1 + x^2y^2}$$

$$\ln kx = \tan^{-1}(xy) \dots (i)$$

passes through (1, 1)

$$\ln k = \frac{\pi}{4} \Rightarrow k = e^{\frac{\pi}{4}}$$

equation (i) becomes

$$\frac{\pi}{4} + \ln x = \tan^{-1}(xy)$$

$$xy = \tan\left(\frac{\pi}{4} + \ln x\right)$$

$$xy = \left(\frac{1 + \tan(\ln x)}{1 - \tan(\ln x)}\right) \dots (ii)$$

put $x = e$ in (ii)

$$\therefore ey(e) = \frac{1 + \tan 1}{1 - \tan 1}$$

Q3 (B)

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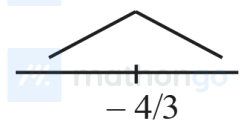
$$(x+1)dy - y dx = e^{3x}(x+1)^2$$

$$\frac{(x+1)dy - ydx}{(x+1)^2} = e^{3x}$$

$$d\left(\frac{y}{x+1}\right) = e^{3x} \Rightarrow \frac{y}{x+1} = \frac{e^{3x}}{3} + C$$

$$\left(0, \frac{1}{3}\right) \Rightarrow C = 0 \Rightarrow y = \frac{(x+1)e^{3x}}{3}$$

$$\frac{dy}{dx} = \frac{1}{3}((x+1)3e^{3x} + e^{3x}) = \frac{3^{3x}}{3}(3x+4)$$



$$-4/3$$

Clearly, $x = -\frac{4}{3}$ is point of local minima

Q4 (C)

$$y^2 dx - xy dy = -(x^2 + y^2) dy$$

$$y(y dx - x dy) = -(x^2 + y^2) dy$$

$$-y(x dx - y dy) = -(x^2 + y^2) dy$$

$$\frac{xdy - ydx}{x^2} = \left(1 + \frac{y^2}{x^2}\right) \frac{dy}{y}$$

$$\Rightarrow \frac{d(y/x)}{1 + \frac{y^2}{x^2}} = \frac{dy}{y}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \ln y + C$$

$$(\alpha, \sqrt{3}\alpha) \Rightarrow \frac{\pi}{3} = \ln(\sqrt{3}\alpha) + \frac{\pi}{4}$$

$$\therefore \ln(\sqrt{3}\alpha) = \frac{\pi}{12}$$

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Q5 (B)

$$\frac{dy}{dx} - \frac{y}{x} = -\frac{3}{2} \left(\frac{y}{x} \right)^2$$

$$y = vx$$

$$\frac{dv}{v^2} = -\frac{3dx}{2x}$$

$$-\frac{1}{v} = -\frac{3}{2} \ln|x| + C$$

$$-\frac{x}{y} = -\frac{3}{2} \ln|x| + C$$

$$x = e, y = \frac{e}{3}$$

$$C = -\frac{3}{2}$$

$$\text{When } x = 1, y = \frac{2}{3}$$

Q6 (12)

$$(4 + x^2)dy - 2x(x^2 + 3y + 4)dx$$

$$(x^2 + 4) \frac{dy}{dx} = 2x^3 + 6xy + 8x$$

$$(x^2 + 4) \frac{dy}{dx} - 6xy = 2x^3 + 8x$$

$$\frac{dy}{dx} - \frac{6x}{x^2 + 4} y = \frac{2x^3 + 8x}{x^2 + 4}$$

$$\text{L.I. } \frac{dy}{dx} + py = \phi$$

$$p = \frac{-6x}{x^2 + 4}$$

$$\phi = \frac{2x^3 + 8x}{x^2 + 4}$$

$$\text{I.F.} = e^{\int \frac{6x}{x^2+4} dx} = e^{-3 \log_e(x^2+4)}$$

Hints and Solutions

MathonGo

$$= e^{\log_e(x^2+4)^{-3}} = \frac{1}{(x^2+4)^3}$$

Sol.

$$y \cdot \frac{1}{(x^2+4)^3} = \int \frac{2x^3+8x}{(x^2+4)^3(x^2+4)} dx$$

$$\frac{y}{(x^2+4)^3} = \int \frac{2x(x^2+4)}{(x^2+4)^3(x^2+4)} dx$$

$$x^2+4 = t$$

$$2x dx = dt$$

$$\frac{y}{(x^2+4)^3} = \int \frac{dt}{t^3}$$

$$\frac{y}{(x^2+4)^3} = \frac{-1}{2(x^2+4)^2} + C$$

passes through origin (0, 0)

$$0 = \frac{-1}{2 \times 16} + C$$

$$\frac{y}{(x^2+4)^3} = \frac{-1}{2(x^2+4)^2} + \frac{1}{32}$$

$$y = \frac{-(x^2+4)}{2} + \frac{(x^2+4)^3}{32}$$

$$y(2) = -8 + \frac{8 \times 8 \times 8}{32}$$

$$y(2) = -8 + 16 = 8$$

Q7 (42)

$$\frac{dy}{dx} = \frac{1}{1 + \sin 2x}$$

$$\int dy = \int \frac{dx}{(\sin x + \cos x)^2}$$

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$$y(x) = -\frac{1}{1 + \tan x} + C$$

$$y\left(\frac{\pi}{4}\right) = \frac{1}{2} = -\frac{1}{2} + C$$

$$C = 1$$

$$y(x) = \frac{-1}{1 + \tan x} + 1$$

$$y(x) = \frac{-1 + 1 + \tan x}{1 + \tan x}$$

$$y(x) = \frac{\tan x}{1 + \tan x}$$

Solving with $y = \sqrt{2} \sin x$

$$\frac{\tan x}{1 + \tan x} = \sqrt{2} \sin x$$

$$\sin x = 0, \quad \frac{1}{\sqrt{2}} = \sin x + \cos x$$

$$x = \pi, \quad \frac{1}{2} = \sin\left(x + \frac{\pi}{4}\right)$$

$$\sin \frac{\pi}{6} = \sin\left(x + \frac{\pi}{4}\right)$$

$$x + \frac{\pi}{4} = \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}$$

$$x = \frac{5\pi}{6} - \frac{\pi}{4}, x = \frac{13\pi}{6} - \frac{\pi}{4}$$

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sum of sol.

$$= \pi + \frac{7\pi}{12} + \frac{23\pi}{12}$$

$$= \frac{12\pi + 7\pi + 23\pi}{12} = \frac{42\pi}{12} = \frac{k\pi}{12}$$

$$\Rightarrow k = 42$$

Q8 (C)

$$I.F. = e^{\int e^x(x^2-2)dx} = e^{\int e^x(x^2-2x+2x-2)dx}$$

$$= e^{e^x(x^2-2x)}$$

$$y \cdot e^{e^x(x^2-2x)} = \int e^{e^x(x^2-2x)} e^x (x^2-2x)(x^2-2) e^x dx$$

$$\text{Let } e^x(x^2-2x) = t$$

$$\text{So, } y \cdot e^{e^x(x^2-2x)} = \int e^t \cdot t \, dt$$

$$\text{At } x = 0, t = 0$$

$$x = 2, t = 0$$

$$= t \cdot e^t - e^t + c$$

$$x = 0; 0 \cdot 1 = 0 - 1 + c \Rightarrow c = 1$$

$$\text{for } x = 2; y \cdot 1 = 0 - 1 + 1 = 0$$

$$y(2) = 0$$

Q9 (B)

Hints and Solutions

MathonGo

Let equation of circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(2x+2g)}{(2y+2f)}$$

Comparing with $\frac{dy}{dx} = \frac{ax - by + a}{bx + cy + a}$

$$\Rightarrow b = 0, a = -2, c = 2$$

$$\Rightarrow -2g = -2 \Rightarrow g = 1 \quad 2f = -2 \\ f = -1$$

Now circle will be

$$x^2 + y^2 + 2x - 2y + c = 0$$

its passes through (2, 5)

which will give $c = -23$

$$\text{so circle will be } x^2 + y^2 + 2x - 2y - 23 = 0$$

centre $C = (-1, 1)$

and radius 5

Now P is (11, 6)

So minimum distance of P from circle will be

$$= \sqrt{(11+1)^2 + (6-1)^2} - 5$$

$$= 13 - 5$$

$$= 8$$

Q10 (D)

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$$\frac{dy}{dx} + \frac{2^{x-y}(2^y-1)}{2^x-1} = 0,$$

$$x, y > 0, y(1) = 1, y(2) = ?$$

$$\frac{dy}{dx} = -\frac{2^x(2^y-1)}{2^y(2^x-1)}$$

$$\int \frac{2^y}{2^y-1} dy = -\int \frac{2^x}{2^x-1} dx$$

$$\frac{1}{\ln 2} \int \frac{2^y \ln 2}{2^y-1} dy = -\frac{1}{\ln 2} \int \frac{2^x \ln 2}{2^x-1} dx$$

$$\frac{1}{\ln 2} \ln|2^y-1| = \frac{-1}{\ln 2} \ln|2^x-1| + C$$

$$\text{At } x=1, y=1$$

Putting this values in above relation we get $C=0$

$$\ln|2^y-1| + \ln|2^x-1| = 0$$

$$(2^x-1)(2^y-1) = 1$$

$$2^y-1 = \frac{1}{2^x-1}$$

$$\text{At } x=2$$

$$2^y = \frac{1}{3} + 1 = \frac{4}{3}$$

$$y = \log_2 \frac{4}{3} = \log_2 4 - \log_2 3 = 2 - \log_2 3$$

Q11 (B)

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2}$$

$$\text{I.f} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$$

$$x e^{\tan^{-1} y} = \int \frac{\tan^{-1} y}{1+y^2} e^{\tan^{-1} y} dy$$

$$x \cdot e^{\tan^{-1} y} = (\tan^{-1} y - 1) e^{\tan^{-1} y} + c$$

$\therefore (1, 0)$ lies on it $c = 2$.

$$\text{For } y = \tan 1 \Rightarrow x = \frac{2}{e}$$

Q12 (320)

$$(1-x^2) \frac{dy}{dx} = xy + (x^3 + 2) \sqrt{1-x^2}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{-x}{1-x^2} \right) y = \frac{x^3 + 2}{\sqrt{1-x^2}}$$

$$\text{I.F} = e^{\int \frac{-x}{1-x^2} dx} = \sqrt{1-x^2}$$

$$y(x) \cdot \sqrt{1-x^2} = \frac{x^4}{4} + 2x + c$$

$$y(0) = 0 \Rightarrow c = 0$$

$$\sqrt{1-x^2} y(x) = \frac{x^4}{4} + 2x$$

$$\text{required value} = \int_{-1/2}^{1/2} \left(\frac{x^4}{4} + 2x \right) dx - \frac{1}{4} \cdot 2 \int_0^{1/2} x^4 dx$$

$$= \frac{1}{10} (x^5)_0^{1/2} = \frac{1}{320}$$

$$k^{-1} = 320$$

Q13 (A)

Hints and Solutions

MathonGo

$$\left(\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}} \right) x \frac{dy}{dx} = x + \left(\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}} \right) y$$

$$\Rightarrow e^{\frac{y}{x}} (x dy - y dx) + \frac{x}{\sqrt{x^2 - y^2}} (x dy - y dx) = x dx$$

Dividing both side by x^2

$$\Rightarrow e^{\frac{y}{x}} \left(\frac{x dy - y dx}{x^2} \right) + \frac{1}{\sqrt{1 - \left(\frac{y}{x} \right)^2}} \left(\frac{x dy - y dx}{x^2} \right) = \frac{dx}{x}$$

$$\Rightarrow e^{\frac{y}{x}} d\left(\frac{y}{x}\right) + \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} d\left(\frac{y}{x}\right) = \frac{dy}{x}$$

Integrate both side.

$$\int e^{\frac{y}{x}} d\left(\frac{y}{x}\right) + \int \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} d\left(\frac{y}{x}\right) = \int \frac{dx}{x}$$

$$\Rightarrow e^{\frac{y}{x}} + \sin^{-1}\left(\frac{y}{x}\right) = \ln x + c$$

It passes through (1, 0)

$$1 + 0 = 0 + c \Rightarrow c = 1$$

It passes through $(2\alpha, \alpha)$

$$e^{\frac{1}{2}} + \sin^{-1} \frac{1}{2} = \ln 2\alpha + 1$$

$$\Rightarrow \ln 2\alpha = \sqrt{e} + \frac{\pi}{6} - 1$$

$$\Rightarrow 2\alpha = e^{\left(\sqrt{e} + \frac{\pi}{6} - 1\right)}$$

$$\Rightarrow \alpha = \frac{1}{2} e^{\left(\frac{\pi}{6} + \sqrt{e} - 1\right)}$$

Q14 (A)

$$x(1 - x^2) \frac{dy}{dx} + (3x^2 y - y - 4x^3) = 0$$

$$x(1 - x^2) \frac{dy}{dx} + (3x^2 - 1)y = 4x^3$$

$$\frac{dy}{dx} + \frac{(3x^2 - 1)}{(x - x^3)} y = \frac{4x^3}{(x - x^3)}$$

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$$\frac{dy}{dx} + Py = Q$$

$$IF = e^{\int P dx} = e^{\int \frac{3x^2-1}{x-x^3} dx}$$

$$x - x^3 = t \Rightarrow IF = e^{\int \frac{-dt}{t}}$$

$$= e^{-\ln t} = \frac{1}{t}$$

$$\therefore IF = \frac{1}{x - x^3}$$

$$y \times IF = \int Q \times IF dx$$

$$y \left(\frac{1}{x - x^3} \right) = \int \frac{4x^3}{x - x^3} \times \frac{1}{(x - x^3)} dx$$

$$= \int \frac{4x^3}{(x - x^3)^2} dx$$

$$= \int \frac{4x}{(1 - x^2)^2} dx \quad 1 - x^2 = K$$

$$= 2 \int \frac{-dK}{K^2} \quad -2x dx = dK$$

$$= -2 \left(-\frac{1}{K} \right) + C$$

$$\frac{y}{x - x^3} = \frac{2}{K} + C$$

$$\frac{y}{x - x^3} = \frac{2}{1 - x^2} + C$$

$$\text{At } x = 2, y = -2$$

$$\frac{-2}{2 - 8} = \frac{2}{1 - 4} + C$$

$$\frac{1}{3} = \frac{-2}{3} + C$$

$$\therefore C = 1$$

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Differential Equations

Put $x = 3$

Hints and Solutions

JEE Main 2022 (June) Chapter-wise Qs Bank

MathonGo

$$\frac{y}{3-27} = \frac{2}{1-9} + 1$$

$$\frac{y}{-24} = -\frac{1}{4} + 1$$

$$\frac{y}{-24} = \frac{3}{4}$$

$$y = \frac{3}{4}(-24) = -18$$

Q15 (D)

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$$2ye^{x/y^2}dx + (y^2 - 4xe^{x/y^2})dy = 0$$

$$2e^{x/y^2} [ydx - 2xdy] + y^2dy = 0$$

$$2e^{x/y^2} \left[\frac{y^2dx - x \cdot (2y)dy}{y} \right] + y^2dy = 0$$

Divide by y^3

$$2e^{x/y^2} \left[\frac{y^2dx - x \cdot (2y)dy}{y^4} \right] + \frac{1}{y}dy = 0$$

$$2e^{x/y^2} d\left(\frac{x}{y^2}\right) + \frac{1}{y}dy = 0$$

Integrating

$$\int 2e^{x/y^2} d\left(\frac{x}{y^2}\right) + \int \frac{1}{y}dy = 0$$

$$2e^{x/y^2} + \ln y + c = 0$$

(0, 1) lies on it.

$$2e^0 + \ln 1 + c = 0 \Rightarrow c = -2$$

Required curve : $\boxed{2e^{x/y^2} + \ln y - 2 = 0}$

For x (e)

$$2e^{x/e^2} + \ln e - 2 = 0 \Rightarrow x = -e^2 \log_e 2$$

Q16 (A)

$$y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{v^2 + 16}$$

$$\Rightarrow \int \frac{dv}{\sqrt{v^2 + 16}} = \int \frac{dx}{x}$$

$$\Rightarrow \ln |v + \sqrt{v^2 + 16}| = \ln x + \ln C$$

$$\Rightarrow y + \sqrt{y^2 + 16x^2} = Cx^2$$

$$\text{As } y(1) = 3 \Rightarrow C = 8$$

$$\Rightarrow y(2) = 15$$

Q17 (2)

Hints and Solutions

MathonGo

$$\frac{dy}{dx} + \frac{\sqrt{2}}{2 \cos^4 x - \cos 2x} y = x e^{\tan^{-1}(\sqrt{2} \cot 2x)}$$

$$\int \frac{dx}{2 \cos^4 x - \cos 2x}$$

$$= \int \frac{dx}{\cos^4 x + \sin^4 x} = \int \frac{\operatorname{cosec}^4 x \, dx}{1 + \cot^4 x}$$

$$= - \int \frac{t^2 + 1}{t^4 + 1} dt = - \int \frac{\left(1 + \frac{1}{t^2}\right)}{\left(t - \frac{1}{t}\right)^2 + 2} dt = \frac{-1}{\sqrt{2}} \tan^{-1} \left(\frac{t - \frac{1}{t}}{\sqrt{2}} \right)$$

$$\cot x = t$$

$$= - \frac{1}{\sqrt{2}} \tan^{-1} (\sqrt{2} \cot 2x)$$

$$\therefore \text{IF} = e^{-\tan^{-1}(\sqrt{2} \cot 2x)}$$

$$y e^{-\tan^{-1}(\sqrt{2} \cot 2x)} = \int x \, dx$$

$$y e^{-\tan^{-1}(\sqrt{2} \cot 2x)} = \frac{x^2}{2} + c$$

$$y \left(\frac{\pi}{4} \right) = \frac{\pi^2}{32} + c \Rightarrow c = 0$$

$$y = \frac{x^2}{2} e^{\tan^{-1}(\sqrt{2} \cot 2x)}$$

$$y \left(\frac{\pi}{3} \right) = \frac{\pi^2}{18} e^{\tan^{-1} \left(\sqrt{2} \cot \frac{2\pi}{3} \right)}$$

$$= \frac{\pi^2}{18} e^{-\tan^{-1} \left(\sqrt{\frac{2}{3}} \right)}$$

$$\alpha = \sqrt{\frac{2}{3}} \Rightarrow 3\alpha^2 = 2$$

Q18 (C)

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$$\frac{dy}{1+y^2} + \frac{2e^x}{1+e^{2x}} dx = 0 \quad (i)$$

on integration

$$\tan^{-1} y + 2 \tan^{-1} e^x = c$$

$$\because y(0) = 0$$

$$\text{so, } C = \frac{\pi}{2} \Rightarrow \tan^{-1} y + 2 \tan^{-1} e^x = \frac{\pi}{2}$$

$$\text{from eq.(i), } \left(\frac{dy}{dx} \right)_{x=0} = -1$$

$$\arg y(\ln \sqrt{3}) = -\frac{1}{\sqrt{3}}$$

$$6 \left[y'(0) + (y(\ln \sqrt{3}))^2 \right] = 6 \left[-1 + \frac{1}{3} \right] = -4$$

Q19 (14)

$$\frac{dy}{dx} + \frac{2x}{x-1} \cdot y = \frac{1}{(x-1)^2}$$

$$y = \frac{1}{(x-1)^2} \left[\frac{e^{2x} + 1}{2e^{2x}} \right]$$

$$y(3) = \frac{e^6 + 1}{8e^6}$$

$$\alpha + \beta = 14$$